IGCSE 0607 Practice Questions — E7.5 — Combined Questions & Markscheme

Non-right-angled triangles

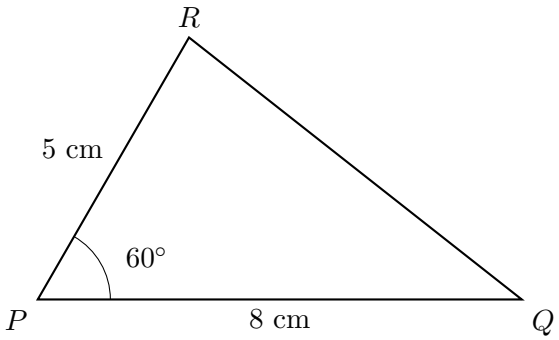
Total Questions: 39 — **Total Marks:** 280 Marks

Table of Contents & Quick Navigation

- **E7.5** Non-right-angled triangles
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(36 Qs)*

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1 The diagram shows triangle PQR.
$PQ = 8$ cm, $PR = 5$ cm and angle $QPR = 60°$.



NOT TO SCALE

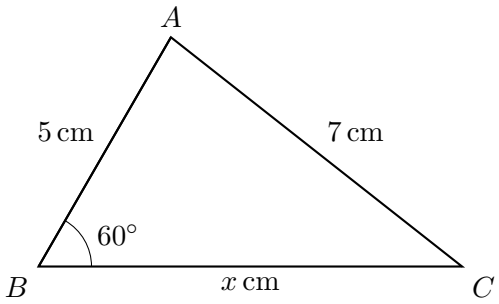
(a) Calculate the exact area of triangle PQR.
Give your answer in the form $a\sqrt{3}$.

..... cm^2 **[2]**

(b) Calculate the exact length of QR.

$QR =$ ..... cm **[3]**

2 In triangle ABC, $AB = 5\,\text{cm}$, $BC = x\,\text{cm}$ and $AC = 7\,\text{cm}$.
Angle $ABC = 60°$.



NOT TO SCALE

(a) Use the cosine rule to show that $x^2 - 5x - 24 = 0$.

[3]

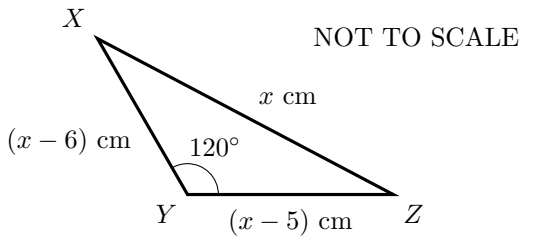
(b) Solve the equation $x^2 - 5x - 24 = 0$ to find the length of BC.

$BC =$ ..... cm **[2]**

(c) Calculate the exact area of triangle ABC.
Give your answer in the form $a\sqrt{b}$, where a and b are integers.

Area = ... cm^2 **[2]**

3 The diagram shows triangle XYZ with $XY = (x - 6)$ cm, $YZ = (x - 5)$ cm, $XZ = x$ cm, and angle $XYZ = 120°$.



(a) Use the cosine rule to show that $2x^2 - 33x + 91 = 0$.

[3]

(b) Solve the equation $2x^2 - 33x + 91 = 0$ and hence write down the lengths of XY, YZ, and XZ.

$XY =$ ..... , $YZ =$ ..... , $XZ =$ ..... **[3]**

(c) Calculate

(i) the exact area of triangle XYZ, giving your answer in the form $k\sqrt{3}$, where k is an integer,

Area = ... cm^2 **[2]**

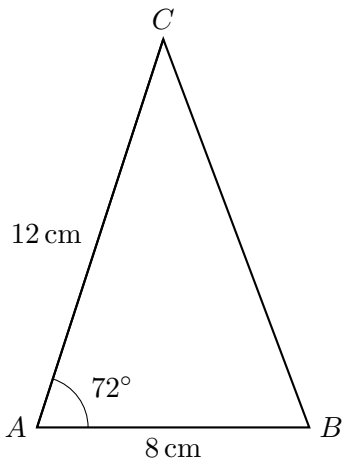
(ii) the exact shortest distance from vertex Y to the side XZ, giving your answer in the form $\frac{p\sqrt{3}}{q}$, where p and q are integers.

Distance = ... cm **[2]**

(d) Find the exact value of $\sin(\angle YXZ)$, giving your answer in the form $\frac{m\sqrt{3}}{n}$, where m and n are integers.

$\sin(\angle YXZ) =$ ..... **[2]**

4 The diagram shows triangle ABC.



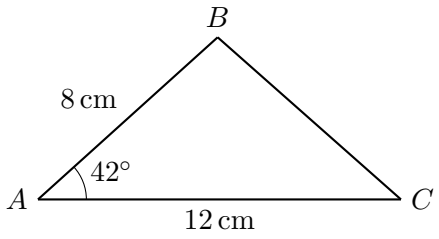
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

5 The diagram shows triangle ABC.
$AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 42°$.



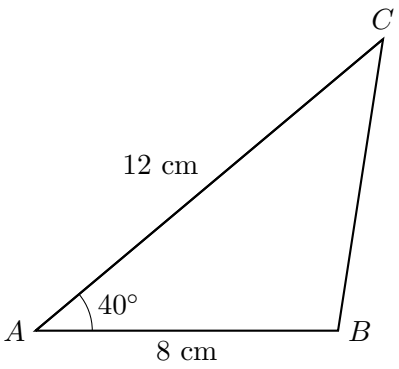
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

$BC =$ ..... cm **[3]**

6 The diagram shows a triangle ABC.



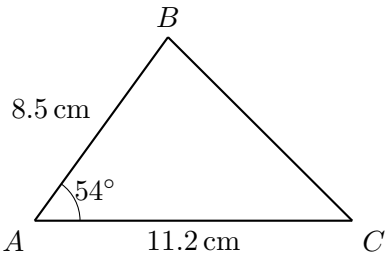
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

7 The diagram shows triangle ABC.



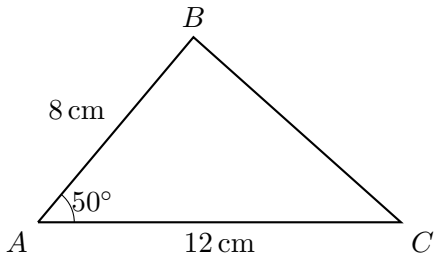
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

8 A triangle ABC is shown below.



NOT TO SCALE

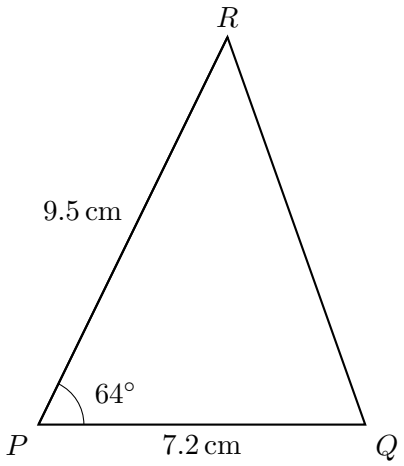
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

9 In triangle PQR, $PQ = 7.2\,\text{cm}$, $PR = 9.5\,\text{cm}$ and angle $QPR = 64°$.



NOT TO SCALE

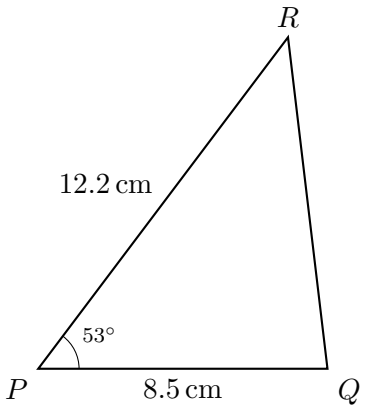
(a) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

(b) Calculate the length of QR.

QR = ... cm **[3]**

10 The diagram shows a triangle PQR.
$PQ = 8.5$ cm, $PR = 12.2$ cm and angle $QPR = 53°$.



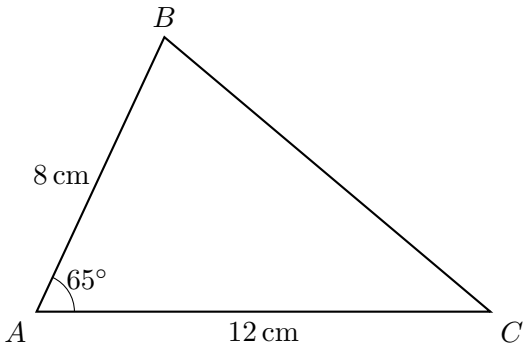
(a) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

(b) Calculate the length of QR.

QR = ... cm **[3]**

11 The diagram shows a triangle ABC with $AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 65°$.



NOT TO SCALE

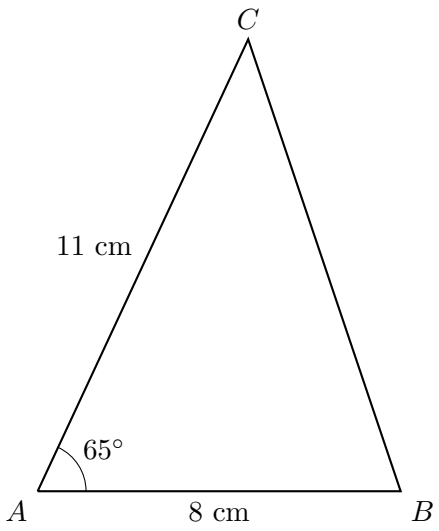
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

12 The diagram shows a triangle ABC.
$AB = 8$ cm, $AC = 11$ cm and angle $BAC = 65°$.



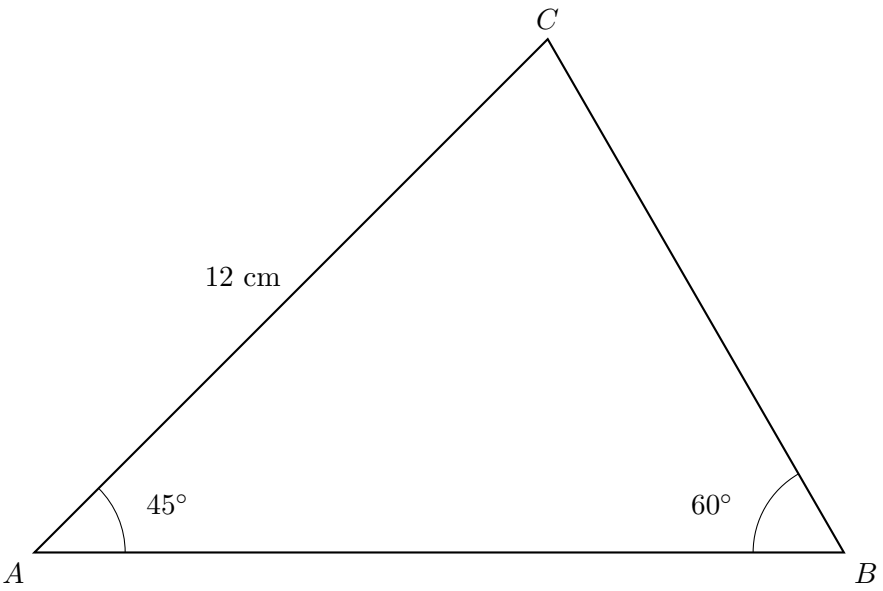
(a) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

(b) Calculate the length of BC.

BC = ... cm **[3]**

13 In triangle ABC, $AC = 12$ cm, angle $BAC = 45°$ and angle $ABC = 60°$.



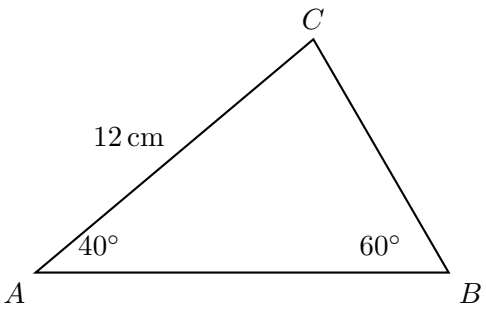
(a) Calculate the length of BC.

..... cm **[3]**

(b) Calculate the area of triangle ABC.

..... cm^2 **[3]**

14 The diagram shows triangle ABC.
$AC = 12$ cm, angle $BAC = 40°$ and angle $ABC = 60°$.



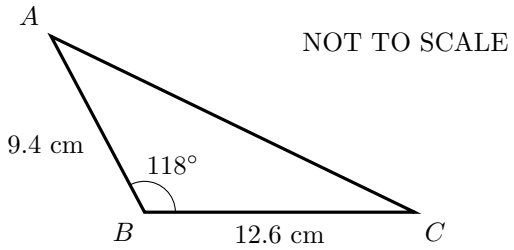
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

15 The diagram shows triangle ABC with $AB = 9.4$ cm, $BC = 12.6$ cm, and obtuse angle $ABC = 118°$.



(a) Calculate the length of AC, giving your answer correct to 3 significant figures.

$AC =$ ..... cm **[3]**

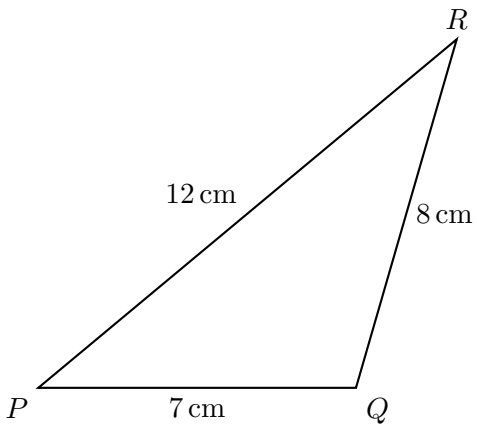
(b) Calculate angle BAC, giving your answer correct to 1 decimal place.

..... ° **[3]**

(c) Calculate the area of triangle ABC, giving your answer correct to 3 significant figures.

Area = ... cm^2 **[2]**

16 The diagram shows triangle PQR.
$PQ = 7$ cm, $QR = 8$ cm and $PR = 12$ cm.



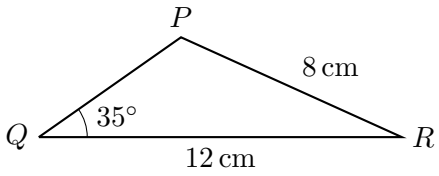
(a) Calculate the largest angle in triangle PQR.

..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

17 In triangle PQR, $PR = 8$ cm, $QR = 12$ cm and angle $PQR = 35°$.
Angle QPR is an obtuse angle.



NOT TO SCALE

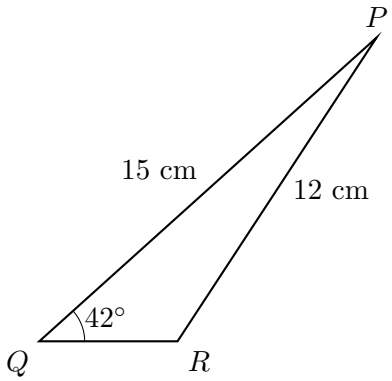
(a) Calculate the size of angle QPR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

18 The diagram shows a triangle PQR.
The length of PQ is 15 cm and the length of PR is 12 cm.
Angle $PQR = 42°$.
Angle PRQ is obtuse.



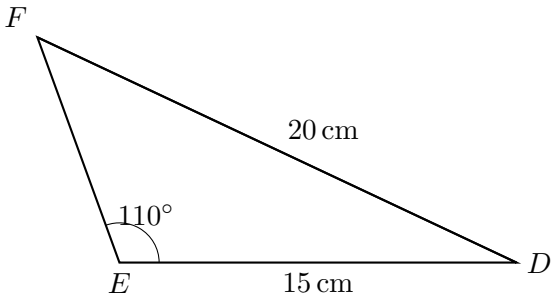
(a) Calculate angle PRQ.

..... ° **[3]**

(b) Calculate the length of QR.

..... cm **[3]**

19 The diagram shows triangle DEF.
$DE = 15$ cm, $DF = 20$ cm, and angle $DEF = 110°$.



NOT TO SCALE

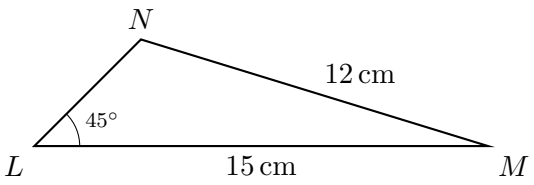
(a) Calculate angle DFE.

Angle $DFE =$ ..... ° **[3]**

(b) Calculate the length of EF.

$EF =$ ..... cm **[3]**

20 The diagram shows a triangle LMN.
$LM = 15$ cm, $MN = 12$ cm and angle $MLN = 45°$.
Angle LNM is obtuse.



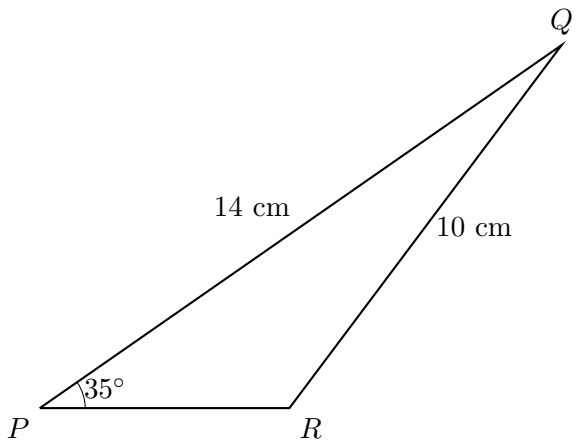
(a) Calculate the size of angle LNM.

Angle LNM = ... ° **[3]**

(b) Calculate the area of triangle LMN.

Area = ... cm^2 **[3]**

21 The diagram shows triangle PQR.
$PQ = 14$ cm, $QR = 10$ cm and angle $QPR = 35°$.
Angle PRQ is obtuse.



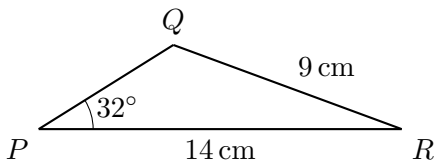
(a) Calculate angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[3]**

22 The diagram shows triangle PQR.
The angle PQR is obtuse.



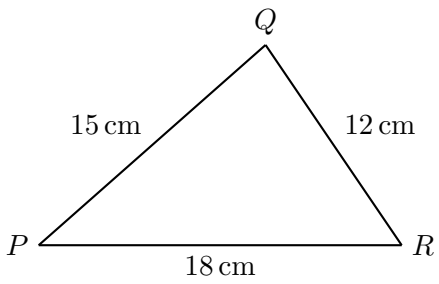
(a) Calculate the obtuse angle PQR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[3]**

23 The diagram shows triangle PQR.
$PQ = 15\,\text{cm}$, $QR = 12\,\text{cm}$ and $PR = 18\,\text{cm}$.



(a) Use the cosine rule to show that angle $PQR = 82.8°$, correct to 1 decimal place.

[3]

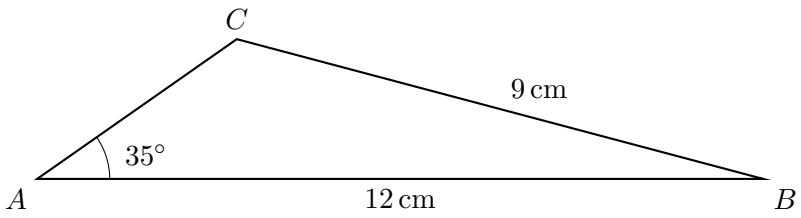
(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

(c) Calculate the shortest distance from R to the line PQ.

..... cm **[2]**

24 In triangle ABC, $AB = 12\,\text{cm}$, $BC = 9\,\text{cm}$ and angle $BAC = 35°$.



NOT TO SCALE

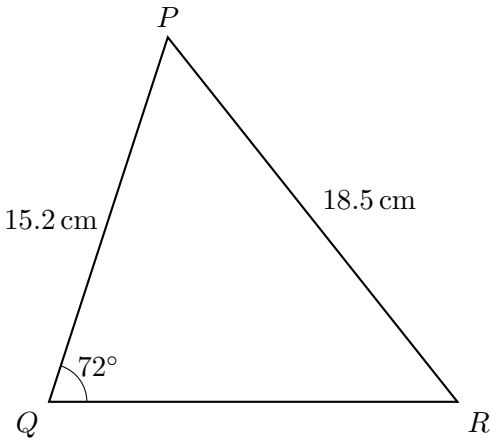
(a) Calculate the two possible values of angle ACB.
Give your answers correct to 1 decimal place.

..... ° **[4]**

(b) Given that angle ACB is obtuse, calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

25 The diagram shows a triangle PQR.
$PQ = 15.2\,\text{cm}$, $PR = 18.5\,\text{cm}$ and angle $PQR = 72°$.



NOT TO SCALE

(a) Calculate the size of angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

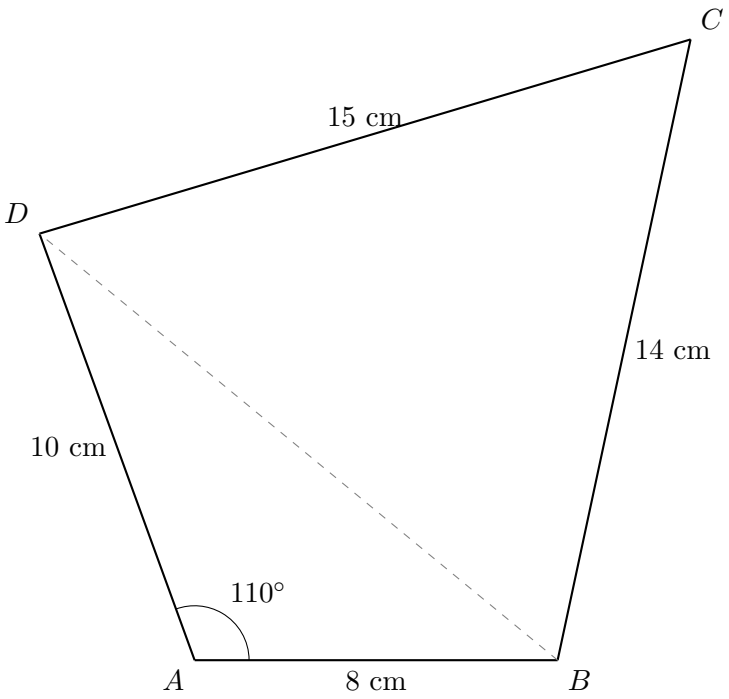
(b) Calculate the length of QR.

$QR =$ ..... cm **[3]**

(c) Calculate the shortest distance from P to the line QR.

Distance = ... cm **[2]**

26 The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $AD = 10$ cm, $BC = 14$ cm and $CD = 15$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of the diagonal BD.

..... cm **[3]**

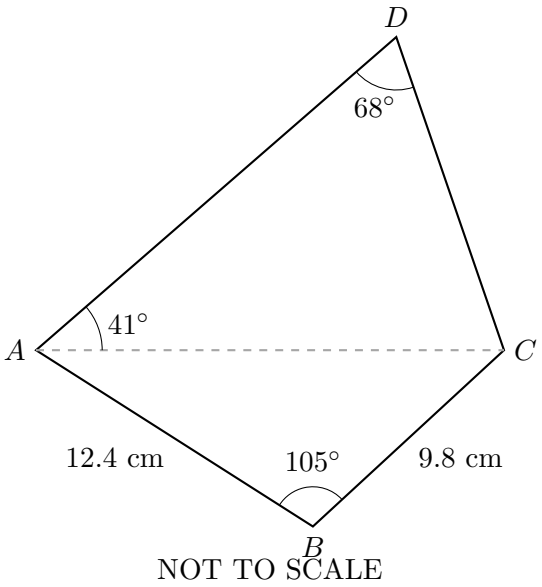
(b) Calculate the angle BCD.

..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

27 The diagram shows a quadrilateral $ABCD$.
$AB = 12.4$ cm, $BC = 9.8$ cm and angle $ABC = 105°$.
Angle $CAD = 41°$ and angle $ADC = 68°$.



NOT TO SCALE

(a) Calculate the length of AC.
Give your answer correct to 3 significant figures.

$AC =$ ..... cm **[3]**

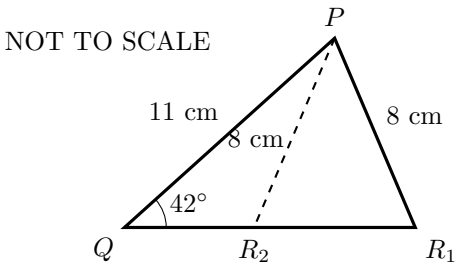
(b) Calculate the length of CD.
Give your answer correct to 3 significant figures.

$CD =$ ..... cm **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

28 In triangle PQR, $PQ = 11$ cm, $PR = 8$ cm, and angle $PQR = 42°$.
As shown in the diagram, there are two possible positions, R_1 and R_2, for vertex R.



(a) Calculate the two possible sizes of angle PRQ, giving your answers correct to 1 decimal place.

Acute angle = ... , Obtuse angle = ... **[4]**

(b) Write down the size of angle QPR_1 and the size of angle QPR_2, giving your answers correct to 1 decimal place.

$Angle QPR_1$ = ... , $Angle QPR_2$ = ... **[2]**

(c) Calculate the difference between the area of triangle PQR_1 and the area of triangle PQR_2, giving your answer correct to 3 significant figures.

Difference = ... cm^2 **[4]**

29 In triangle ABC, $AB = 15$ cm, $BC = 12$ cm and angle $BAC = 48°$.

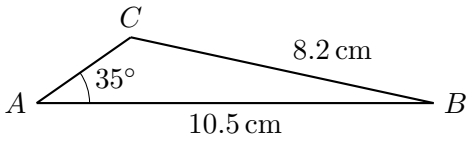
(a) Calculate the two possible values of angle BCA.

and ..... **[3]**

(b) Hence, calculate the difference between the two possible areas of triangle ABC.

..... cm^2 **[4]**

30 In triangle ABC, $AB = 10.5\,\text{cm}$, $BC = 8.2\,\text{cm}$ and angle $BAC = 35°$.
Angle BCA is obtuse.



(a) Calculate the size of angle BCA.

..... ° **[3]**

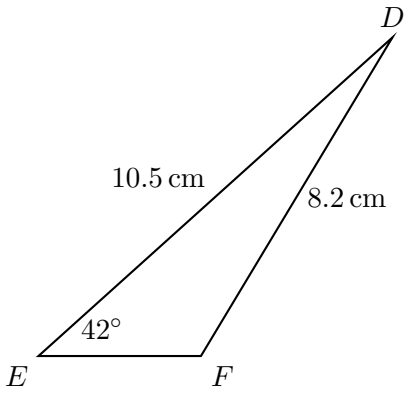
(b) Calculate the length of AC.

$AC =$ ..... cm **[3]**

(c) A point D lies outside the triangle ABC such that $CD = 6.4\,\text{cm}$ and angle ACD is acute.
The area of triangle ACD is $8.5\,\text{cm}^2$.
Calculate the size of angle ACD.

..... ° **[2]**

31 The diagram shows triangle DEF.
$DE = 10.5$ cm, $DF = 8.2$ cm and angle $DEF = 42°$.
Angle DFE is an obtuse angle.



(a) Calculate angle DFE.

Angle $DFE =$ ..... ° **[3]**

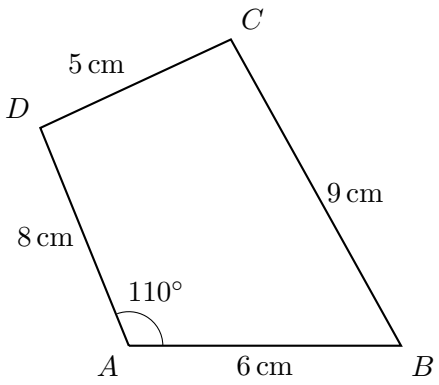
(b) Calculate the length of EF.

$EF =$ ..... cm **[3]**

(c) Calculate the shortest distance from D to the line EF.

..... cm **[2]**

32 The diagram shows a quadrilateral $ABCD$.
$AB = 6$ cm, $AD = 8$ cm, $DC = 5$ cm, and $BC = 9$ cm.
Angle $DAB = 110°$.



NOT TO SCALE

(a) Calculate the length of the diagonal BD.

..... *cm* **[3]**

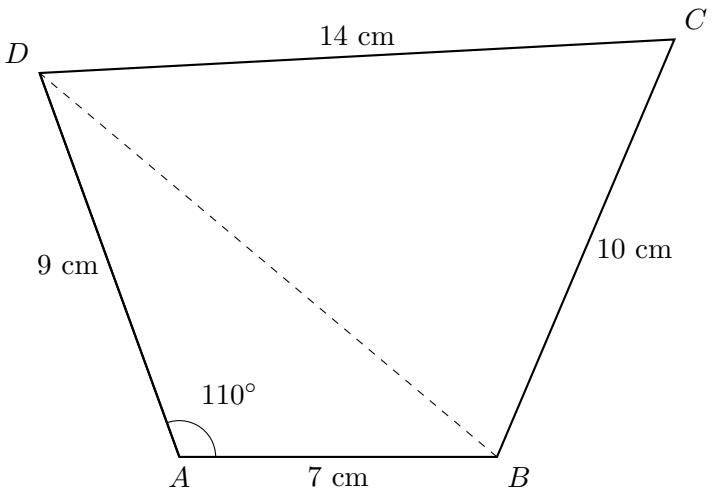
(b) Calculate the size of angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

33 The diagram shows a quadrilateral $ABCD$.
$AB = 7$ cm, $AD = 9$ cm, $BC = 10$ cm and $CD = 14$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of BD.

..... cm **[3]**

(b) Calculate angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

34 The diagram shows a quadrilateral $ABCD$.
$AB = 7.4$ cm, $BC = 9.1$ cm, and $CD = 10.5$ cm.
Angle $ABC = 112°$ and angle $CAD = 43°$.

(a) Calculate the length of AC.

..... cm **[3]**

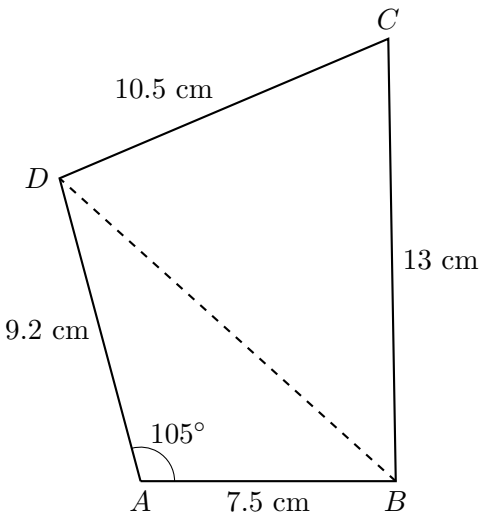
(b) Calculate the acute angle ADC.

..... ° **[3]**

(c) Calculate the shortest distance from D to the line AC.

..... cm **[3]**

35 The diagram shows a quadrilateral $ABCD$.
$AB = 7.5$ cm, $AD = 9.2$ cm and angle $BAD = 105°$.
$BC = 13$ cm and $CD = 10.5$ cm.



(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

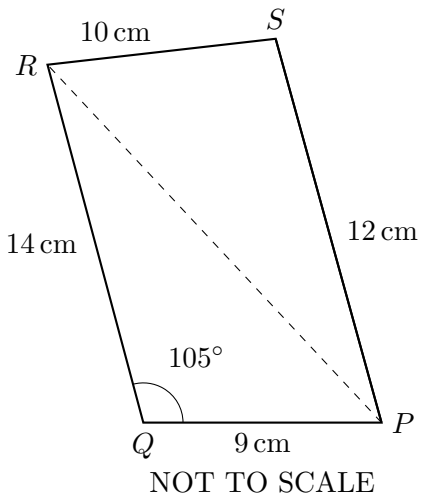
(b) Calculate angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[3]**

36 The diagram shows a quadrilateral $PQRS$.
$PQ = 9$ cm, $QR = 14$ cm, and angle $PQR = 105°$.
$PS = 12$ cm and $RS = 10$ cm.



NOT TO SCALE

(a) Calculate the length of the diagonal PR.

$PR =$ ..... cm **[3]**

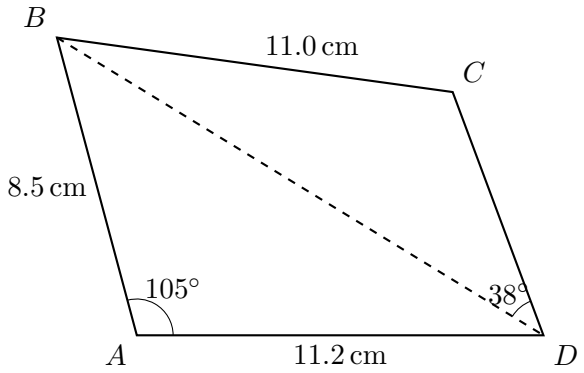
(b) Calculate angle PSR.

Angle $PSR =$ ..... ° **[4]**

(c) Calculate the total area of the quadrilateral $PQRS$.

Area = ... cm^2 **[3]**

37 The diagram shows a quadrilateral $ABCD$.
$AB = 8.5\,\text{cm}$, $AD = 11.2\,\text{cm}$ and angle $BAD = 105°$.
$BC = 11.0\,\text{cm}$ and angle $BDC = 38°$.
Angle BCD is obtuse.



NOT TO SCALE

(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

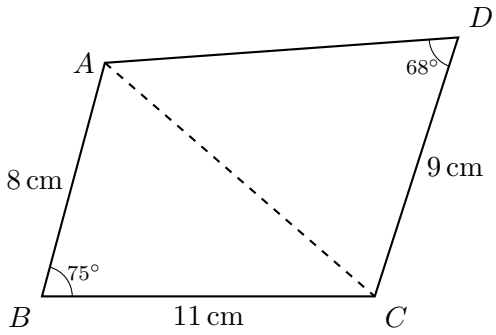
(b) Calculate the size of angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[4]**

38 The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $BC = 11$ cm and $CD = 9$ cm.
Angle $ABC = 75°$ and angle $ADC = 68°$.



(a) Calculate the length of AC.

$AC =$ ..... cm **[3]**

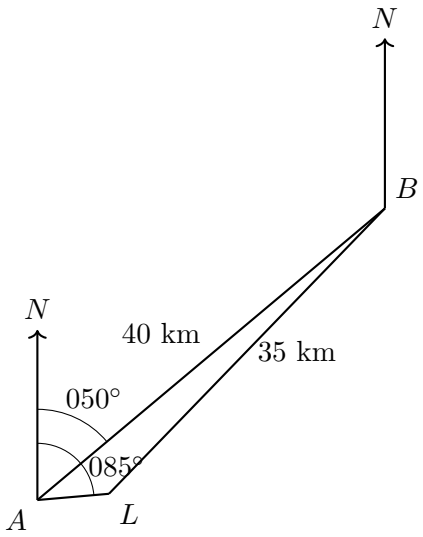
(b) Calculate the size of angle CAD.

Angle $CAD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[4]**

39 Ship A and Ship B are at sea.
Ship B is 40 km from Ship A on a bearing of 050°.
A lighthouse L is 35 km from Ship B.
The bearing of L from A is 085°.
Given that the angle ALB is obtuse:



(a) Calculate the obtuse angle ALB.

..... ° **[4]**

(b) Calculate the distance AL.

..... km **[3]**

(c) Calculate the bearing of L from B.

..... ° **[3]**

1 **(a)** $\frac{1}{2} \times 8 \times 5 \times \sin 60°$ **[M1]**

$10\sqrt{3}$ **[A1]**

(b) $QR^2 = 8^2 + 5^2 - 2(8)(5)\cos 60°$ **[M1]**

$QR^2 = 64 + 25 - 80\left(\frac{1}{2}\right)$ **[M1]**

7 **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** $7^2 = 5^2 + x^2 - 2(5)(x)\cos 60°$ **[M1]**

$\cos 60° = \frac{1}{2}$ or 0.5 seen **[B1]**

$49 = 25 + x^2 - 5x \implies x^2 - 5x - 24 = 0$ (with no errors seen) **[A1]**

(b) $(x - 8)(x + 3) = 0$ **[M1]**

$x = 8$ **[A1]**

(c) Area $= \frac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

3 **(a)** $x^2 = (x - 6)^2 + (x - 5)^2 - 2(x - 6)(x - 5)\cos 120°$ **[M1]**

Using $\cos 120° = -\frac{1}{2}$: $x^2 = (x^2 - 12x + 36) + (x^2 - 10x + 25) + (x^2 - 11x + 30)$ **[M1]**

$x^2 = 3x^2 - 33x + 91$ simplifying to $2x^2 - 33x + 91 = 0$ **[A1]**

(b) $(2x - 7)(x - 13) = 0$ **[M1]**

$x = 13$ (rejecting $x = \frac{7}{2}$ as side lengths must be positive) **[A1]**

$XY = 7$ cm, $YZ = 8$ cm, $XZ = 13$ cm **[A1]**

(c) **(i)** Area $= \frac{1}{2} \times 7 \times 8 \times \sin 120° = 28 \times \frac{\sqrt{3}}{2}$ **[M1]**

$14\sqrt{3}$ **[A1]**

(ii) $\frac{1}{2} \times 13 \times d = 14\sqrt{3}$ **[M1]**

$\frac{28\sqrt{3}}{13}$ **[A1]**

(d) $\frac{\sin(\angle YXZ)}{8} = \frac{\sin 120°}{13}$ giving $\sin(\angle YXZ) = \frac{8 \times \frac{\sqrt{3}}{2}}{13}$ **[M1]**

$\frac{4\sqrt{3}}{13}$ **[A1]**

[Back to Question] *[Total Marks: 12]*

4 **(a)** $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos 72°$ **[M1]**

$BC^2 = 208 - 59.33\ldots$ OR $148.66\ldots$ **[M1]**

12.2 (accept $12.19\ldots$) **[A1]**

(b) Area $= \frac{1}{2}(8)(12)\sin 72°$ **[M1]**

45.7 (accept $45.65\ldots$) **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $0.5 \times 8 \times 12 \times \sin(42°)$ **[M1]**

32.1 (or 32.11 to 32.12) **[A1]**

(b) $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(42°)$ **[M1]**

$BC^2 = 65.3...$ **[M1]**

8.08 (or 8.081 to 8.082) **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin 40°$ **[M1]**

30.9 (accept $30.85\ldots$) **[A1]**

(b) $8^2 + 12^2 - 2 \times 8 \times 12 \times \cos 40°$ **[M1]**

$60.9\ldots$ **[A1]**

7.80 or 7.81 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** $0.5 \times 8.5 \times 11.2 \times \sin(54°)$ **[M1]**

38.5 (38.50 to 38.51) **[A1]**

(b) $BC^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos(54°)$ [M2]
(or [M1] for correct cosine rule formula with one substitution error)

$BC^2 = 85.77...$ **[A1]**

9.26 (9.261...) **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin(50°)$ **[M1]**

36.77... so 36.8 **[A1]**

(b) $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(50°)$ **[M1]**

$BC^2 = 208 - 123.415... = 84.58...$ **[M1]**

9.197... so 9.20 **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** $\frac{1}{2} \times 7.2 \times 9.5 \times \sin 64°$ **[M1]**

30.7 or 30.73... **[A1]**

(b) $7.2^2 + 9.5^2 - 2(7.2)(9.5)\cos 64°$ **[M2]**

(or $M1$ for correct implicit cosine rule equation)

9.06 or 9.062... **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** Uses Area $= \frac{1}{2} \times 8.5 \times 12.2 \times \sin 53°$ **[M1]**

41.4 (or 41.40 to 41.41) **[A1]**

(b) Uses cosine rule: $QR^2 = 8.5^2 + 12.2^2 - 2 \times 8.5 \times 12.2 \times \cos 53°$ **[M1]**

$QR^2 = 72.25 + 148.84 - 207.4 \times \cos 53° = 96.27...$ **[A1]**

9.81 **[A1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** Uses cosine rule: $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos 65°$ **[M1]**

$BC^2 = 126.85...$ **[M1]**

11.3 **[A1]**

(b) Uses area formula: $\frac{1}{2} \times 8 \times 12 \times \sin 65°$ **[M1]**

43.5 **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** $\frac{1}{2} \times 8 \times 11 \times \sin(65°)$ **[M1]**

39.9 **[A1]**

(b) $BC^2 = 8^2 + 11^2 - 2(8)(11)\cos(65°)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

10.5 **[A1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** $\frac{BC}{\sin 45°} = \frac{12}{\sin 60°}$ **[M1]**

$BC = \frac{12 \sin 45°}{\sin 60°}$ **[M1]**

9.80 **[A1]**

(b) Angle $ACB = 180° - (45° + 60°) = 75°$ **[B1]**

Area $= \frac{1}{2} \times 12 \times 9.798 \times \sin 75°$ **[M1]**

56.8 **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** $\frac{BC}{\sin 40°} = \frac{12}{\sin 60°}$ **[M1]**

$BC = \frac{12 \sin 40°}{\sin 60°}$ **[M1]**

8.91 (accept 8.907 . . .) **[A1]**

(b) Angle $BCA = 180° - (40° + 60°) = 80°$ **[B1]**

$\frac{1}{2} \times 12 \times 8.91 \times \sin 80°$ **[M1]**

52.6 (accept 52.62 . . . to 52.65 . . .) **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** $AC^2 = 9.4^2 + 12.6^2 - 2 \times 9.4 \times 12.6 \times \cos 118°$ **[M1]**

$AC^2 = 88.36 + 158.76 - (-111.208 \ldots) = 358.328 \ldots$ **[M1]**

18.9 **[A1]**

(b) $\frac{\sin \angle BAC}{12.6} = \frac{\sin 118°}{18.929 \ldots}$ **[M1]**

$\sin \angle BAC = \frac{12.6 \sin 118°}{18.929 \ldots} = 0.5877 \ldots$ **[M1]**

36.0 **[A1]**

(c) Area $= \frac{1}{2} \times 9.4 \times 12.6 \times \sin 118°$ $(= 52.288 \ldots)$ **[M1]**

52.3 **[A1]**

[Back to Question] *[Total Marks: 8]*

16 **(a)** $\cos Q = \frac{7^2 + 8^2 - 12^2}{2 \times 7 \times 8}$ [M2] (M1 for correct implicit cosine rule equation e.g. $12^2 = 7^2 + 8^2 - 2(7)(8) \cos Q$)

106.1 (accept 106.07 . . .) **[A1]**

(b) $\frac{1}{2} \times 7 \times 8 \times \sin 106.1°$ **[M1]**

26.9 (accept 26.89 . . .) **[A1]**

[Back to Question] *[Total Marks: 5]*

17 **(a)** $\dfrac{\sin P}{12} = \dfrac{\sin 35°}{8}$ **[M1]**

$\sin P = \dfrac{12 \sin 35°}{8} = 0.8603...$ **[M1]**

Acute angle $= 59.35...°$ **[A1]**

Obtuse angle $P = 180° - 59.35...° = 120.64...°$ so $120.6°$ **[A1]**

(b) Angle $R = 180° - 35° - 120.64...° = 24.36...°$ **[M1]**

Area $= \dfrac{1}{2} \times 8 \times 12 \times \sin(24.36...°) = 19.79...$ so 19.8 **[A1]**

[Back to Question] *[Total Marks: 6]*

18 **(a)** $\dfrac{\sin R}{15} = \dfrac{\sin 42°}{12}$ **[M1]**

$\sin R = \dfrac{15 \sin 42°}{12}$ (evaluates to $0.836\ldots$) **[M1]**

123.2 (accept 123.2 to 123.3) **[A1]**

(b) Angle $P = 180° - 42° - 123.24° = 14.76°\ldots$ **[M1]**

$\dfrac{QR}{\sin 14.76°} = \dfrac{12}{\sin 42°}$ (or equivalent application of sine/cosine rule) **[M1]**

4.57 (accept 4.56 to 4.58) **[A1]**

[Back to Question] *[Total Marks: 6]*

19 **(a)** $\dfrac{\sin F}{15} = \dfrac{\sin 110°}{20}$ **[M1]**

$\sin F = \dfrac{15 \sin 110°}{20}$ OR $0.7047\ldots$ **[M1]**

44.8 (accept 44.80 to 44.81) **[A1]**

(b) Angle $D = 180° - 110° - 44.81° = 25.19°$ **[B1]**

$\dfrac{EF}{\sin 25.19°} = \dfrac{20}{\sin 110°}$ OR $EF^2 = 15^2 + 20^2 - 2(15)(20)\cos 25.19°$ **[M1]**

9.06 (accept 9.05 to 9.06) **[A1]**

[Back to Question] *[Total Marks: 6]*

20 **(a)** Uses sine rule: $\dfrac{\sin(\text{angle } LNM)}{15} = \dfrac{\sin 45°}{12}$ **[M1]**

$\sin(\text{angle } LNM) = 0.8838...$ leading to acute angle $62.11...°$ **[M1]**

117.9 (or 117.88 to 117.89) **[A1]**

(b) Finds angle $LMN = 180° - 45° - 117.89° = 17.11°$ **[B1]**

Uses Area $= \dfrac{1}{2} \times 15 \times 12 \times \sin 17.11°$ **[M1]**

26.5 (or 26.47 to 26.49) **[A1]**

[Back to Question] *[Total Marks: 6]*

21 **(a)** $\dfrac{\sin R}{14} = \dfrac{\sin 35°}{10}$ **[M1]**

$\sin^{-1}\left(\dfrac{14\sin 35°}{10}\right)$ leading to $53.4°$ $(53.41...)$ **[M1]**

126.6 $(126.58...)$ **[A1]**

(b) Angle $PQR = 180° - 35° - 126.58° = 18.4°$ **[M1]**

Area $= \dfrac{1}{2} \times 14 \times 10 \times \sin(18.4°)$ **[M1]**

22.1 $(22.11...)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

22 **(a)** $\dfrac{\sin Q}{14} = \dfrac{\sin 32°}{9}$ **[M1]**

$\sin Q = \dfrac{14\sin 32°}{9}$ or $0.824...$ **[M1]**

Acute angle $= 55.5°$ $(55.51...)$ **[A1]**

$180 - 55.5... = 124.5°$ $(124.48$ to $124.5)$ **[A1]**

(b) Angle $PRQ = 180 - 32 - 124.48 = 23.52°$ **[B1]**

$0.5 \times 14 \times 9 \times \sin(23.52°)$ **[M1]**

25.1 $(25.12$ to $25.14)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

23 **(a)** $\cos Q = \dfrac{15^2 + 12^2 - 18^2}{2 \times 15 \times 12}$ **[M1]**

$\cos Q = \dfrac{45}{360}$ OR 0.125 **[M1]**

$Q = \cos^{-1}(0.125) = 82.819...°$ **[A1]**

(b) $0.5 \times 15 \times 12 \times \sin(82.8°)$ **[M1]**

89.3 (or 89.28 to 89.30) **[A1]**

(c) Area $= 0.5 \times 15 \times h = 89.3$ OR $h = 12 \times \sin(82.8°)$ **[M1]**

11.9 (or 11.90 to 11.91) **[A1]**

[Back to Question] *[Total Marks: 7]*

24 **(a)** $\dfrac{\sin C}{12} = \dfrac{\sin 35°}{9}$ **[M1]**

$\sin C = \dfrac{12\sin 35°}{9}$ $(= 0.764...)$ **[M1]**

$49.9°$ (or $49.88...$) **[A1]**

$130.1°$ (or $130.11...$) **[A1]**

(b) Angle $ABC = 180 - (35 + 130.1) = 14.9°$ **[M1]**

Area $= \frac{1}{2} \times 12 \times 9 \times \sin(14.9°)$ **[M1]**

13.9 (accept 13.87... to 13.89...) **[A1]**

[Back to Question] *[Total Marks: 7]*

25 **(a)** Uses sine rule: $\frac{\sin R}{15.2} = \frac{\sin 72°}{18.5}$ **[M1]**

$\sin R = 0.781...$ **[M1]**

51.4 **[A1]**

(b) Angle $P = 180° - 72° - 51.38° = 56.62°$ **[B1]**

$\frac{QR}{\sin 56.62°} = \frac{18.5}{\sin 72°}$ **[M1]**

16.2 **[A1]**

(c) Shortest distance $= 15.2 \times \sin 72°$ **[M1]**

14.5 **[A1]**

[Back to Question] *[Total Marks: 8]*

26 **(a)** $BD^2 = 8^2 + 10^2 - 2(8)(10)\cos 110°$ **[M1]**

$BD^2 = 164 - 160(-0.342) = 218.72$ **[M1]**

14.8 **[A1]**

(b) $\cos C = \frac{14^2 + 15^2 - 14.79^2}{2(14)(15)}$ **[M1]**

$\cos C = \frac{202.26}{420} = 0.4816$ **[M1]**

61.2 **[A1]**

(c) Area $ABD = \frac{1}{2}(8)(10)\sin 110° = 37.59$ **[M1]**

Area $BCD = \frac{1}{2}(14)(15)\sin 61.2° = 92.01$ **[M1]**

130 (or 129.6) **[A1]**

[Back to Question] *[Total Marks: 9]*

27 **(a)** $AC^2 = 12.4^2 + 9.8^2 - 2(12.4)(9.8)\cos 105°$ **[M1]**

$AC^2 = 153.76 + 96.04 - (-62.90)$ or 312.7 **[M1]**

17.7 (accept 17.68 . . .) **[A1]**

(b) $\frac{CD}{\sin 41°} = \frac{17.68}{\sin 68°}$ **[M1]**

$CD = \frac{17.68 \times \sin 41°}{\sin 68°}$ **[M1]**

12.5 (accept 12.51 . . .) **[A1]**

(c) Area $ABC = \frac{1}{2} \times 12.4 \times 9.8 \times \sin 105° = 58.68\ldots$ **[B1]**

Area $ACD = \frac{1}{2} \times 17.68 \times 12.51 \times \sin(180° - 41° - 68°) = 104.6\ldots$ (accept correct alternative using $AD = 18.0$) **[M1]**

163 (accept 163.2 to 163.3) **[A1]**

[Back to Question] *[Total Marks: 9]*

28 **(a)** $\frac{\sin \angle PRQ}{11} = \frac{\sin 42°}{8}$ **[M1]**

$\sin \angle PRQ = \frac{11 \sin 42°}{8} = 0.92005\ldots$ **[M1]**

Acute angle $= 66.9°$ **[A1]**

Obtuse angle $= 180° - 66.934\ldots° = 113.1°$ **[A1]**

(b) Angle $QPR_1 = 180° - 42° - 66.934\ldots° = 71.1°$ **[B1]**

Angle $QPR_2 = 180° - 42° - 113.066\ldots° = 24.9°$ **[B1]**

(c) Area of triangle $PQR_1 = \frac{1}{2} \times 11 \times 8 \times \sin 71.066\ldots° = 41.619\ldots\ cm^2$ **[M1]**

Area of triangle $PQR_2 = \frac{1}{2} \times 11 \times 8 \times \sin 24.934\ldots° = 18.549\ldots\ cm^2$ **[M1]**

Difference $= 41.619\cdots - 18.549\cdots = 23.070\ldots$ **[M1]**

23.1 **[A1]**

[Back to Question] *[Total Marks: 10]*

29 **(a)** $\frac{\sin C}{15} = \frac{\sin 48°}{12}$ **[M1]**

$\sin C = 0.9289\ldots$ **[A1]**

68.3 and 111.7 **[A1]**

(b) Calculates either possible angle B: $180° - 48° - 68.27° = 63.73°$ OR $180° - 48° - 111.73° = 20.27°$ **[M1]**

Area $1 = \frac{1}{2} \times 15 \times 12 \times \sin 63.73° = 80.7\ldots$ **[M1]**

Area $2 = \frac{1}{2} \times 15 \times 12 \times \sin 20.27° = 31.2\ldots$ **[M1]**

49.5 (accept 49.5 to 49.6) **[A1]**

[Back to Question] *[Total Marks: 7]*

30 **(a)** $\frac{\sin C}{10.5} = \frac{\sin 35°}{8.2}$ **[M1]**

$\sin C = \frac{10.5 \times \sin 35°}{8.2}$ OR 0.734... **[M1]**

$C = 180° - 47.3° = 132.7°$ (or 132.73 to 132.74) **[A1]**

(b) Angle $B = 180° - 35° - 132.7° = 12.3°$ **[M1]**

$\dfrac{AC}{\sin 12.3°} = \dfrac{8.2}{\sin 35°}$ OR $AC^2 = 10.5^2 + 8.2^2 - 2(10.5)(8.2)\cos 12.3°$ **[M1]**

3.04 (or 3.035 to 3.045) **[A1]**

(c) $0.5 \times 3.04 \times 6.4 \times \sin(ACD) = 8.5$ **[M1]**

$\sin(ACD) = \dfrac{8.5}{9.728}$ OR Angle $ACD = \sin^{-1}\left(\dfrac{8.5}{9.728}\right) = 60.9°$ (or 60.8... to 61.1°) **[A1]**

[Back to Question] *[Total Marks: 8]*

31 **(a)** $\dfrac{\sin F}{10.5} = \dfrac{\sin 42°}{8.2}$ **[M1]**

$\sin F = 0.8568\ldots$ yielding acute angle $58.96\ldots°$ **[A1]**

$180° - 58.96° = 121.0$ (accept $121.04\ldots$) **[A1]**

(b) Angle $EDF = 180° - 42° - 121.04° = 16.96°$ **[B1]**

$\dfrac{EF}{\sin 16.96°} = \dfrac{8.2}{\sin 42°}$ **[M1]**

3.57 (accept $3.56\ldots$ to $3.58\ldots$) **[A1]**

(c) Shortest distance $= 10.5 \times \sin 42°$ **[M1]**

7.03 (accept $7.025\ldots$ to $7.027\ldots$) **[A1]**

[Back to Question] *[Total Marks: 8]*

32 **(a)** $BD^2 = 6^2 + 8^2 - 2(6)(8)\cos(110°)$ **[M1]**

$BD^2 = 100 - 96(-0.342...) = 132.83...$ **[M1]**

$BD = 11.52...$ so 11.5 **[A1]**

(b) $\cos C = \dfrac{9^2 + 5^2 - (11.525...)^2}{2(9)(5)}$ **[M1]**

$\cos C = \dfrac{106 - 132.83...}{90} = -0.2981...$ **[M1]**

$C = 107.34...°$ so 107.3° **[A1]**

(c) Area of triangle $ABD = \dfrac{1}{2} \times 6 \times 8 \times \sin(110°) = 22.55...$ **[M1]**

Area of triangle $BCD = \dfrac{1}{2} \times 9 \times 5 \times \sin(107.34...°) = 21.47...$ **[M1]**

Total Area $= 22.55... + 21.47... = 44.02...$ so 44.0 **[A1]**

[Back to Question] *[Total Marks: 9]*

33 **(a)** $7^2 + 9^2 - 2 \times 7 \times 9 \times \cos 110°$ **[M1]**

$173.09\ldots$ **[A1]**

13.2 (accept 13.15 to 13.2) **[A1]**

(b) $13.156^2 = 10^2 + 14^2 - 2 \times 10 \times 14 \times \cos(\angle BCD)$ **[M1]**

$\cos(\angle BCD) = \dfrac{10^2 + 14^2 - 173.09}{2 \times 10 \times 14}$ **[M1]**

64.0 (accept 63.9 . . . to 64.0) **[A1]**

(c) $\frac{1}{2} \times 7 \times 9 \times \sin 110°$ (evaluates to 29.6 . . .) **[M1]**

$\frac{1}{2} \times 10 \times 14 \times \sin 64.0°$ (evaluates to 62.9 . . .) **[M1]**

92.5 (accept 92.4 to 92.6) **[A1]**

[Back to Question] *[Total Marks: 9]*

34 **(a)** $AC^2 = 7.4^2 + 9.1^2 - 2(7.4)(9.1)\cos(112°)$ [M2]
(or [M1] for correct cosine rule formula with one error)

$AC = 13.7$ (13.71 to 13.72) **[A1]**

(b) $\dfrac{\sin D}{13.712} = \dfrac{\sin 43°}{10.5}$ **[M1]**

$\sin D = \dfrac{13.712 \sin 43°}{10.5}$ or 0.890... **[M1]**

63.0 (62.95 to 63.0) **[A1]**

(c) Angle $ACD = 180 - 43 - 62.95 = 74.05°$ **[M1]**

Shortest distance $= 10.5 \times \sin(74.05°)$ **[M1]**

10.1 (10.09 to 10.1) **[A1]**

[Back to Question] *[Total Marks: 9]*

35 **(a)** $BD^2 = 7.5^2 + 9.2^2 - 2(7.5)(9.2)\cos(105°)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

13.3 (13.28...) **[A1]**

(b) $\cos C = \dfrac{13^2 + 10.5^2 - 13.289^2}{2(13)(10.5)}$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

67.9 (67.91...) **[A1]**

(c) Area $ABD = \frac{1}{2} \times 7.5 \times 9.2 \times \sin(105°)$ (33.32...) **[M1]**

Area $BCD = \frac{1}{2} \times 13 \times 10.5 \times \sin(67.9°)$ (63.24...) **[M1]**

96.6 (96.5... to 96.6) **[A1]**

[Back to Question] *[Total Marks: 9]*

36 **(a)** $PR^2 = 9^2 + 14^2 - 2(9)(14)\cos 105°$ **[M1]**

$PR^2 = 277 - (-65.22 \ldots)$ OR $342.22 \ldots$ **[M1]**

18.5 (accept 18.49 to 18.50) **[A1]**

(b) Identifies correct use of cosine rule for angle PSR: $\cos S = \dfrac{12^2 + 10^2 - PR^2}{2(12)(10)}$ **[M1]**

$\cos S = \dfrac{144 + 100 - 342.22}{240}$ **[M1]**

$\cos S = -0.409\ldots$ **[M1]**

114.2 (accept 114.1 to 114.2) **[A1]**

(c) Area of triangle $PQR = \dfrac{1}{2}(9)(14)\sin 105°$ OR $60.85\ldots$ **[M1]**

Area of triangle $PRS = \dfrac{1}{2}(10)(12)\sin 114.16°$ OR $54.74\ldots$ **[M1]**

Total Area $= 60.85 + 54.74 = 115.6$ (accept 115.5 to 115.6) **[A1]**

[Back to Question] *[Total Marks: 10]*

37 (a) Uses cosine rule: $BD^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos 105°$ **[M1]**

$BD^2 = 246.96...$ **[A1]**

15.7 **[A1]**

(b) Uses sine rule: $\dfrac{\sin C}{15.715} = \dfrac{\sin 38°}{11.0}$ **[M1]**

Acute angle $C = 61.59...°$ **[M1]**

Obtuse angle $BCD = 180° - 61.59...° = 118.4°$ **[A1]**

(c) Area of triangle $ABD = \dfrac{1}{2} \times 8.5 \times 11.2 \times \sin 105°$ $(= 45.97...)$ **[M1]**

Angle $DBC = 180° - 38° - 118.41° = 23.59°$ **[B1]**

Area of triangle $BCD = \dfrac{1}{2} \times 11.0 \times 15.715 \times \sin 23.59°$ $(= 34.59...)$ **[M1]**

80.6 **[A1]**

[Back to Question] *[Total Marks: 10]*

38 (a) Uses cosine rule: $AC^2 = 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 75°$ **[M1]**

$AC^2 = 64 + 121 - 176 \times 0.2588... = 139.448...$ **[A1]**

11.8 (or 11.80 to 11.81) **[A1]**

(b) Uses sine rule in triangle ADC: $\dfrac{\sin(\text{angle } CAD)}{9} = \dfrac{\sin 68°}{11.81}$ **[M1]**

$\sin(\text{angle } CAD) = \dfrac{9 \times \sin 68°}{11.81} = 0.7066...$ **[M1]**

45.0 (or 44.95 to 44.96) **[A1]**

(c) Calculates area of triangle $ABC = \dfrac{1}{2} \times 8 \times 11 \times \sin 75° = 42.50...$ **[M1]**

Finds angle $ACD = 180° - 68° - 44.96° = 67.04°$ **[M1]**

Calculates area of triangle $ADC = \dfrac{1}{2} \times 11.81 \times 9 \times \sin 67.04° = 48.93...$ **[M1]**

Total Area $= 42.50 + 48.93 = 91.4$ (or 91.42 to 91.44) **[A1]**

[Back to Question] *[Total Marks: 10]*

39 **(a)** Angle $BAL = 085° - 050° = 35°$ **[B1]**

$\dfrac{\sin(ALB)}{40} = \dfrac{\sin 35°}{35}$ **[M1]**

$\sin(ALB) = \dfrac{40 \sin 35°}{35} = 0.6555\ldots$ **[M1]**

Acute angle $= 40.96°$, Obtuse angle $ALB = 180° - 40.96° = 139.0$ **[A1]**

(b) Angle $ABL = 180° - 35° - 139.04° = 5.96°$ **[B1]**

$\dfrac{AL}{\sin 5.96°} = \dfrac{35}{\sin 35°}$ **[M1]**

6.33 **[A1]**

(c) Bearing of A from $B = 180° + 050° = 230°$ **[M1]**

Bearing of L from $B = 230° -$ Angle $ABL = 230° - 5.96°$ **[M1]**

224.0 **[A1]**

[Back to Question] *[Total Marks: 10]*